

Supplement C — Extended Collective Dynamics in Kuramoto and Stuart–Landau Systems

C1. Purpose and Scope

This supplement extends the collective dynamics analysis presented in the main manuscript. The objective is to examine synchronization organization, metastability, adaptive recovery, hysteresis behavior, memory persistence, and topology-dependent organization in Kuramoto and Stuart–Landau oscillator systems.

C2. Kuramoto Synchronization Organization

The Kuramoto regime extends the framework toward collective synchronization dynamics. Intermediate coupling regions displayed adaptive transition behavior associated with partial synchronization, transition stabilization, and adaptive organization (see Figures C1–C3).

C3. Topology Robustness

Additional scans explored topology-dependent synchronization behavior across alternative network structures. The observed synchronization organization signatures remained detectable across multiple topological regimes, although quantitative behavior varied between architectures (see Figures C2 and C10).

C4. Stuart–Landau Hysteresis

The Stuart–Landau regime introduced adaptive oscillator dynamics with hysteresis-dependent adaptive transition behavior. The resulting organization patterns demonstrated memory-sensitive transitions and delayed synchronization recovery (see Figure C4).

C5. Metastability Landscapes

Metastability maps revealed intermediate regions characterized by fluctuating yet persistent organization states. These regions were associated with temporary coexistence between partially synchronized and desynchronized dynamics (see Figure C5).

C6. Recovery Dynamics

Recovery analyses demonstrated delayed return behavior near adaptive transition regions. The observed recovery patterns support the interpretation that compatibility-dependent organization may temporarily stabilize adaptive collective structures (see Figure C6).

C7. Adaptive Coupling and Memory Effects

Adaptive coupling scans revealed persistent organization traces after perturbation events. Persistent memory-lock behavior emerged near the adaptive transition edge. Memory-lock refers to prolonged retention of synchronization structure after perturbation events. Memory-sensitive adaptive transition behavior appeared in both synchronization recovery and cluster persistence metrics (see Figures C7 and C9).

C8. Outsider and Cluster Dynamics

Additional analyses explored local synchronization disruption and outsider-induced organization effects. Localized perturbations generated transient synchronization fragmentation before global stabilization was restored (see Figure C8).

C9. Interpretation Boundaries

The supplement does not claim equivalence between oscillator systems or universal synchronization laws. The presented evidence should instead be interpreted as exploratory comparative support for recurring collective organization signatures across adaptive nonlinear systems.

C10. Quantitative Comparative Summary

Table C1. Qualitative comparative summary across adaptive oscillator systems.

Subsystem	Observed signature	Robustness behavior	Interpretation
Kuramoto	Synchronization organization	Stable across topology scans	Collective adaptive organization
Kuramoto	Transition persistence	Reduced after perturbation	Intermediate coupling stabilization
Stuart–Landau	Metastability	Persistent across adaptive scans	Temporary collective stabilization
Stuart–Landau	Recovery delay	Localized near adaptive transition regions	Memory-sensitive organization
Adaptive coupling	Persistent organization traces	Recovered after stabilization	Adaptive collective memory

C10A. Quantitative Collective Metrics

Table C2 provides representative quantitative coupling regions and persistence metrics corresponding to the qualitative comparative summary presented in Table C1.

Representative collective-dynamics scans revealed intermediate synchronization regions associated with metastability, delayed recovery, adaptive memory persistence, and topology-dependent organization across Kuramoto and Stuart–Landau oscillator systems.

Table C2. Quantitative collective metrics across Kuramoto and Stuart–Landau systems.

Subsystem	Representative region	Recovery / memory behavior	Topology robustness	Interpretation
Kuramoto	$K \approx 0.70\text{--}0.875$	persistent synchronization transition	stable across topology scans	collective adaptive organization
Kuramoto	$K \approx 0.80$ peak region	maximal synchronization organization response	robust across alternative topologies	intermediate coupling stabilization
Stuart–Landau	$K \approx 0.45\text{--}0.70$	metastability and hysteresis persistence	stable under adaptive scans	temporary collective stabilization
Adaptive coupling	memory-lock edge near $K \approx 0.70$	persistent recovery traces	retained after perturbation	adaptive collective memory

Representative synchronization organization landscapes and topology-dependent adaptive transitions are illustrated in Figures C1–C10.

The collective-dynamics layer therefore functions as an exploratory comparative synchronization framework rather than a proof of universal collective behavior.

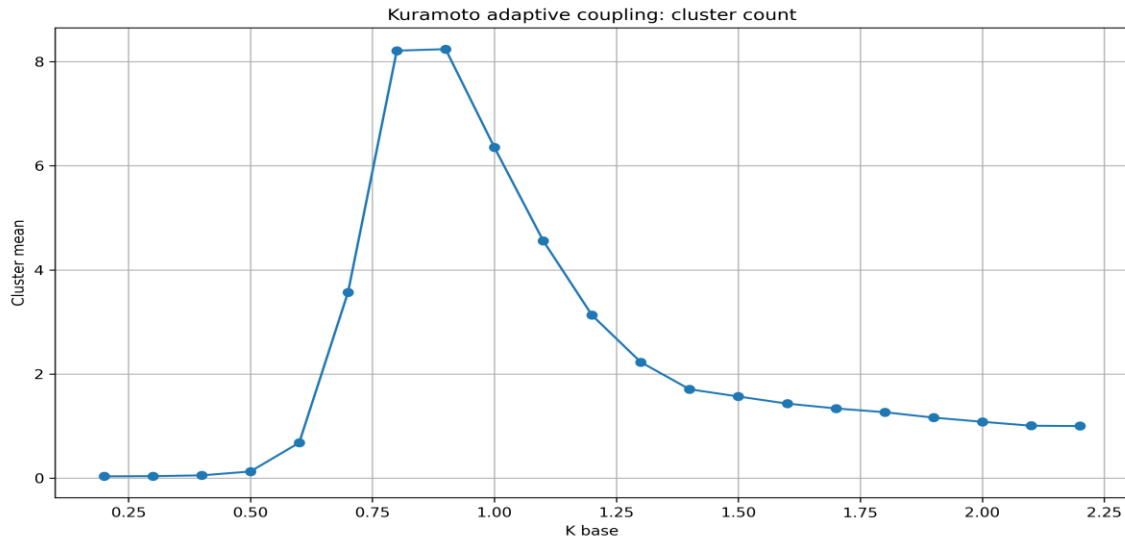


Figure C1. Representative synchronization organization landscape in the Kuramoto regime.

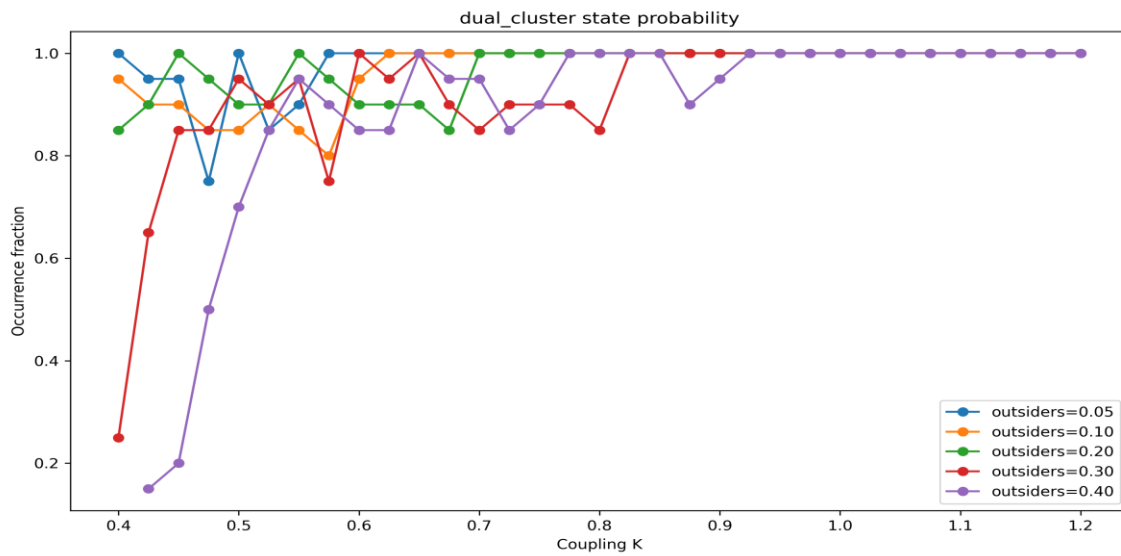


Figure C2. Topology-dependent synchronization robustness across alternative coupling structures.

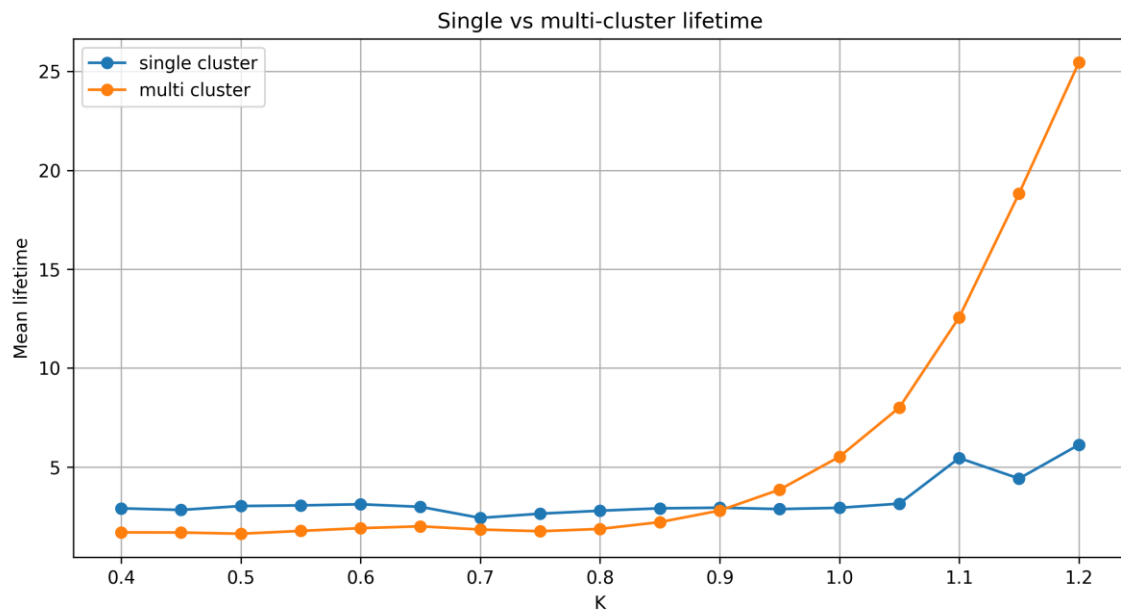


Figure C3. Intermediate synchronization transition region associated with adaptive transition behavior.

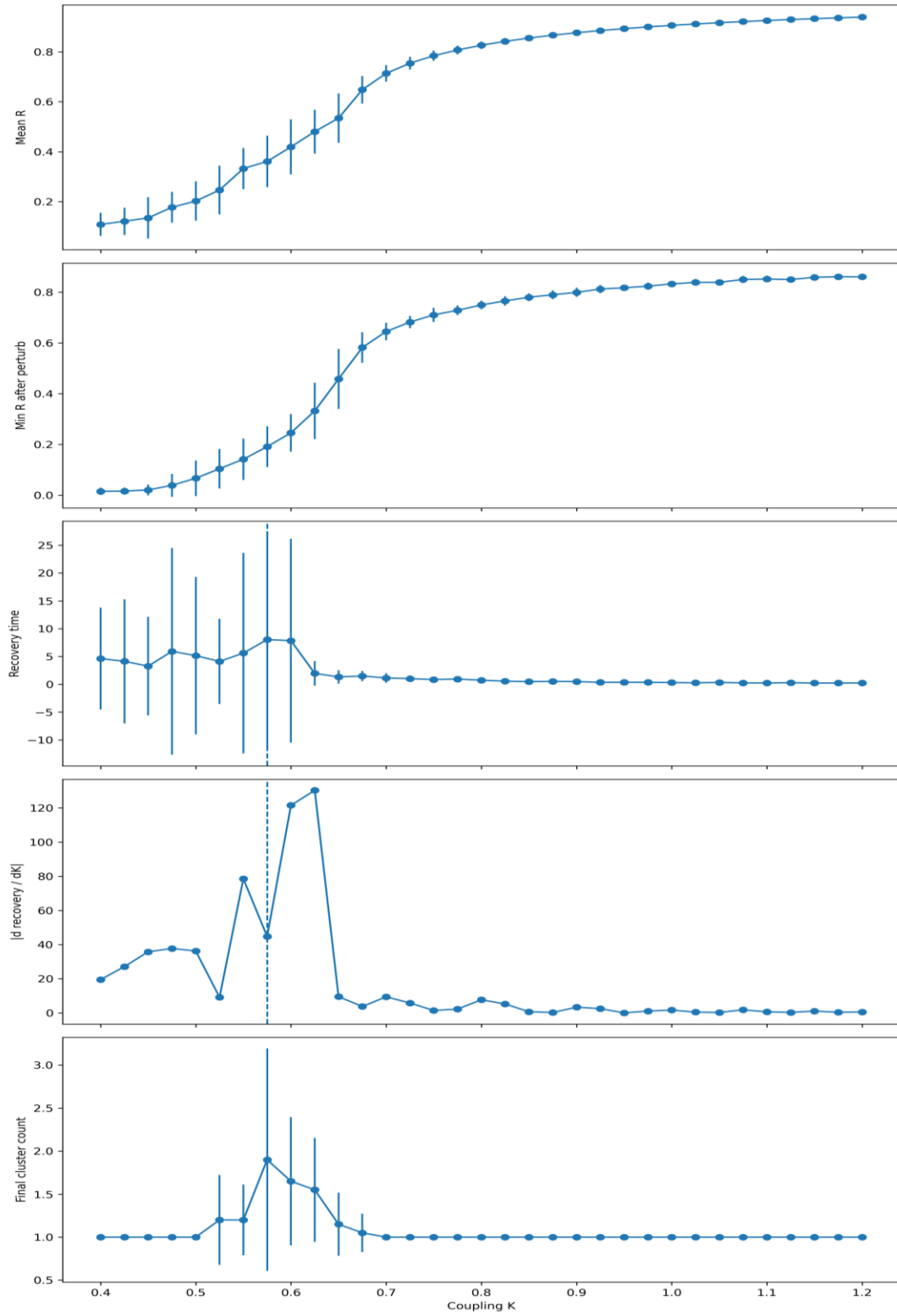


Figure C4. Hysteresis loop structure in the Stuart–Landau adaptive oscillator regime.

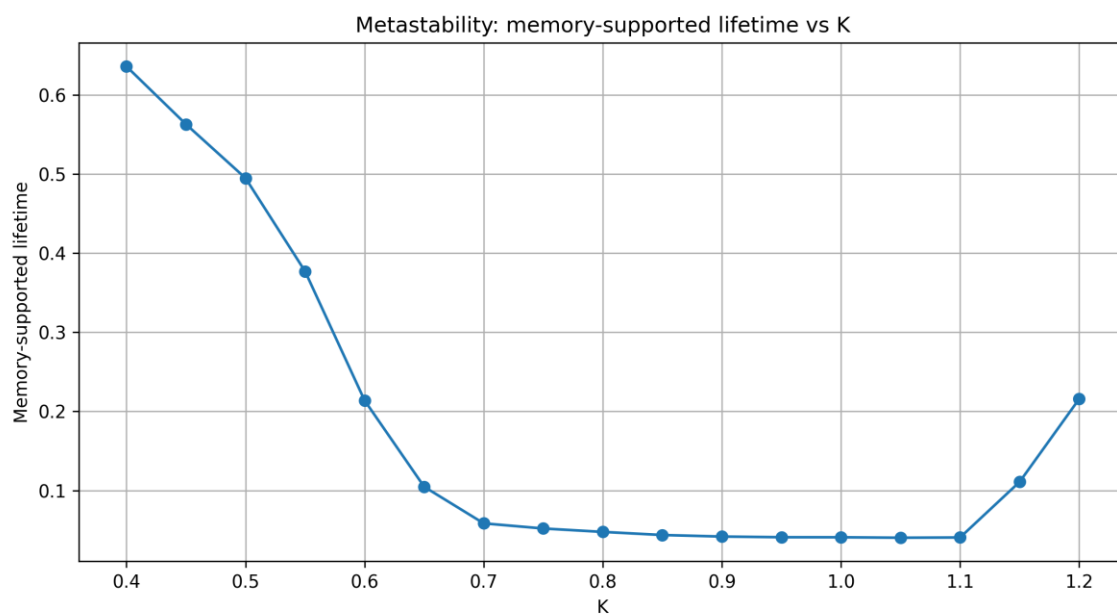


Figure C5. Metastability landscape showing temporary coexistence of partially synchronized states.

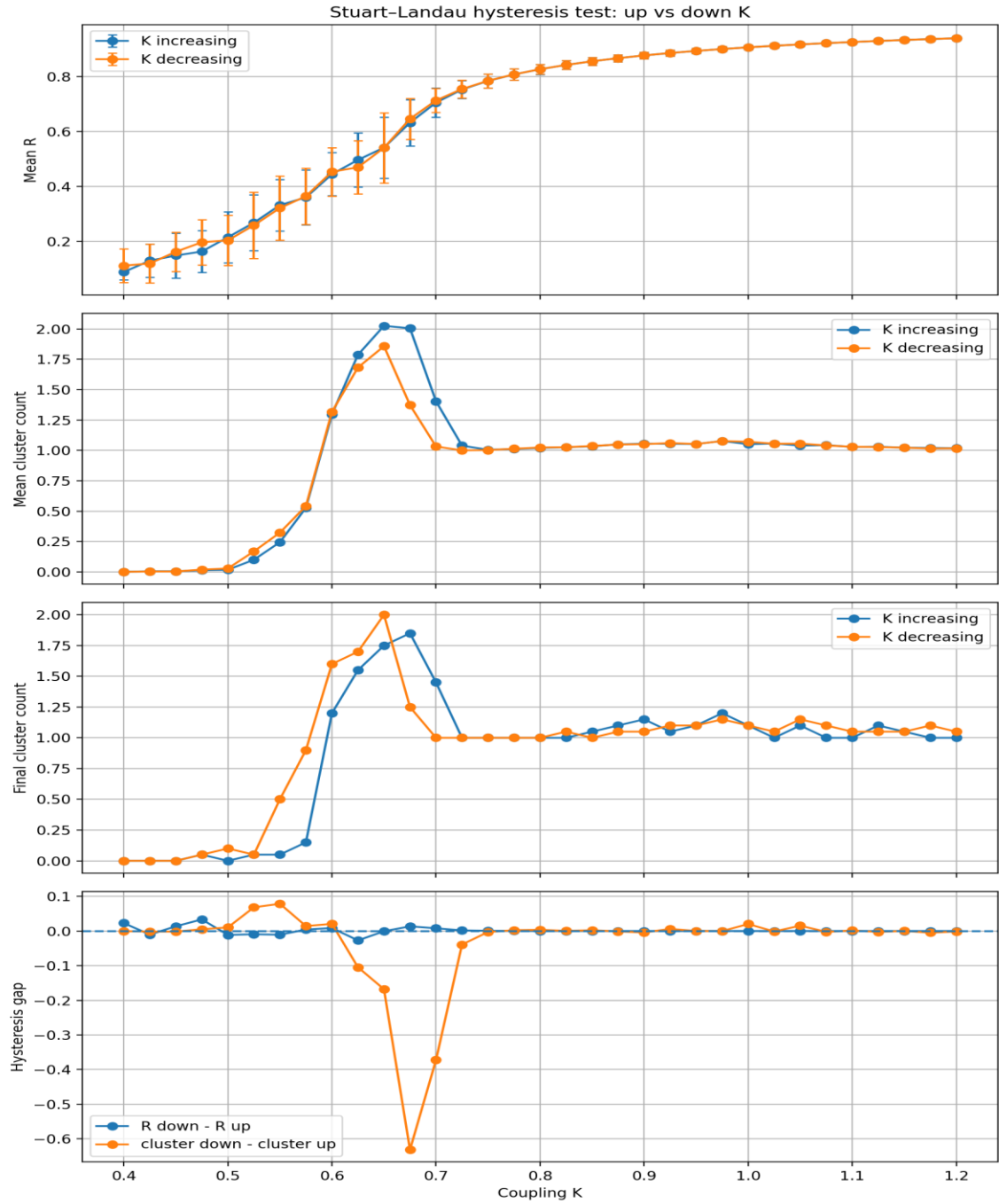


Figure C6. Recovery dynamics near adaptive transition regions after perturbation events.

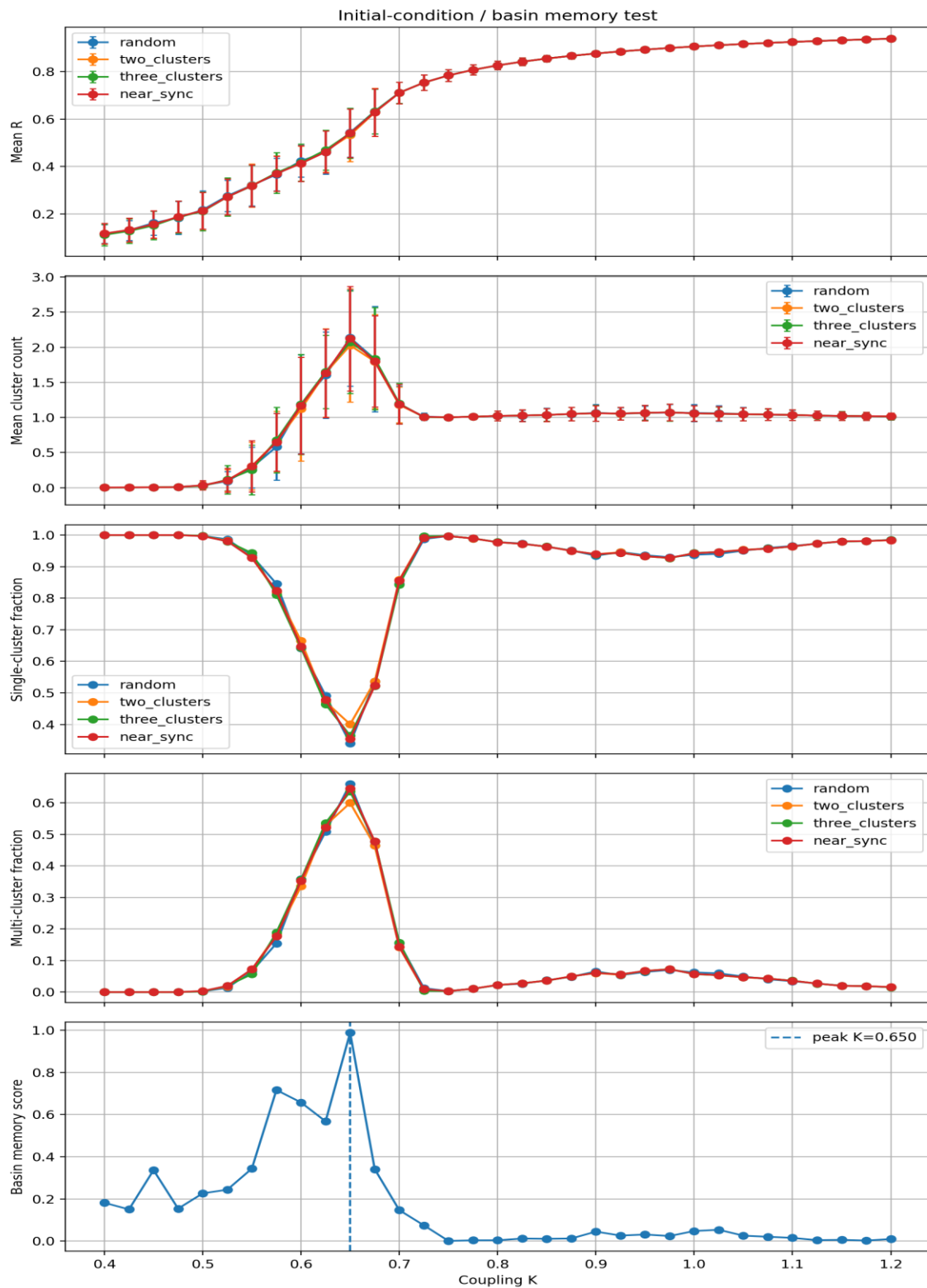


Figure C7. Adaptive coupling scan illustrating persistent organization traces.

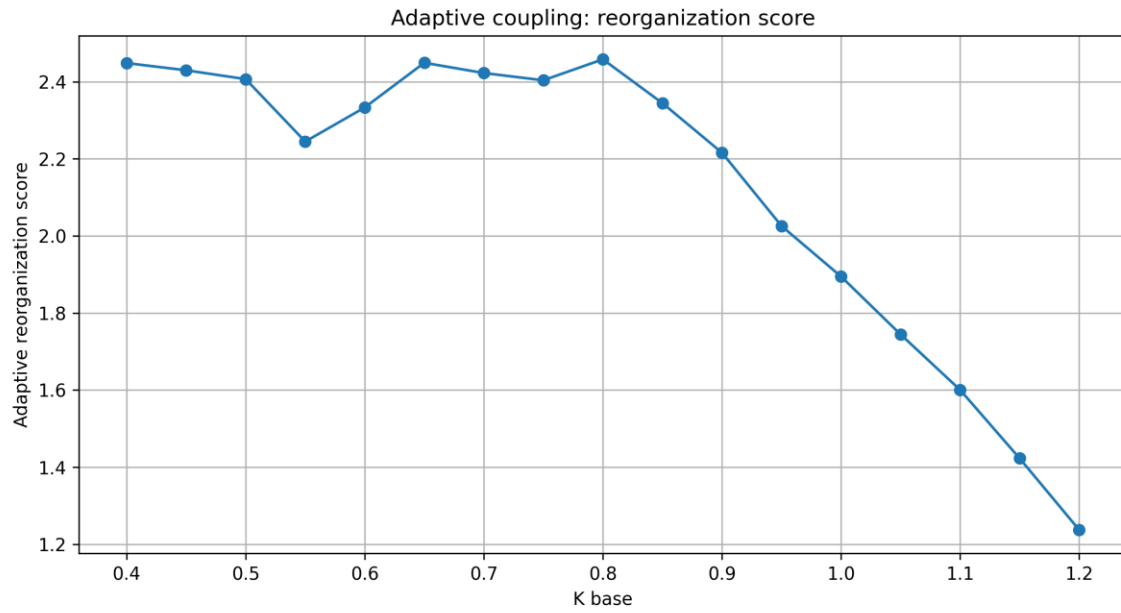


Figure C8. Cluster persistence and synchronization fragmentation behavior.

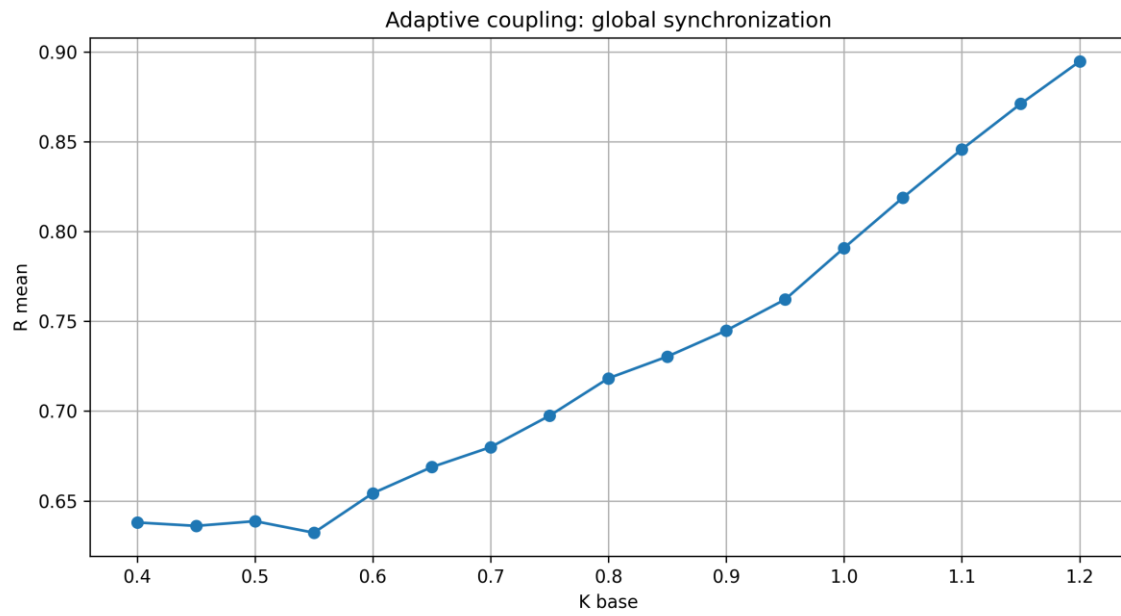


Figure C9. Memory-sensitive organization response under repeated perturbation.

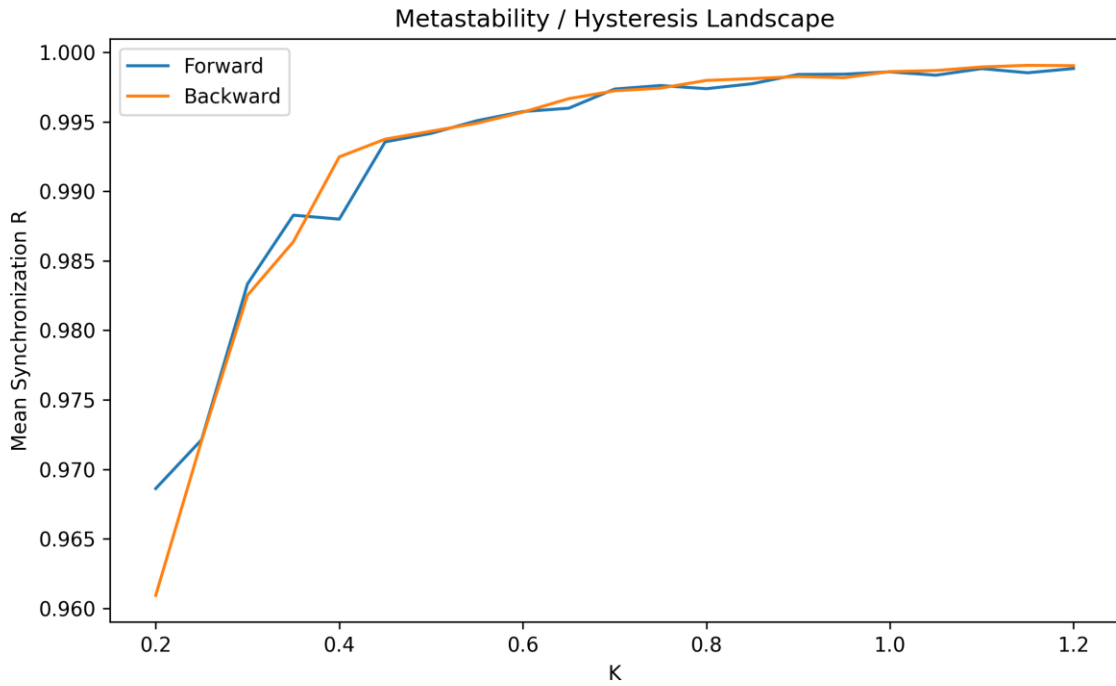


Figure C10. Comparative synchronization persistence under alternative topology conditions.

C11. Supplementary Closing Statement

The present supplement extends the synchronization organization and adaptive collective dynamics layers of the main manuscript. The assembled evidence supports the interpretation that compatibility-dependent organization can emerge in adaptive oscillator systems under intermediate coupling and adaptive transition regimes.